

Soykot Podder

#Machine Learning Engineer, ANTT Robotics, #Dhaka, #Bangladesh

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RESEARCH INTERESTS

#Machine Learning(ML), #Deep Learning(DL), #Large Language Model(LLM), #Cyber Physical System, #Hardware Security

EDUCATION

Indian Institute of Information Technology(IIITDMJ), #Jabalpur, #India #Sep-29 #2021 — #Sep-30 #2023
Master of Science in Artificial Intelligence and Machine Learning Cumulative GPA: 7.9/10.00
Thesis Title: Hardware Trojan detection Using Ensemble learning model

#**American International University of Bangladesh(AIUB)**, #Dhaka, #Bangladesh #March-01 #2016 — #May-30 #2020
Bachelor of Science in Computer Science and Engineering Cumulative GPA: 3.06/4.00
#Major field: Computer Vision, Human computer interaction.

ACADEMIC EXPERIENCE

ANTT Robotics #Dhaka, #Bangladesh
Level-1 Machine Learning Engineer #Feb-01 #2024 — Present

- # Designed, developed, and optimized machine learning models for real-time applications, including predictive maintenance, vehicle detection, and market price forecasting.
- # Conducted data collection, preprocessing, and feature engineering to ensure high-quality input for model training and evaluation.
- # Deployed machine learning models in production environments, ensuring scalability and performance using APIs and cloud-based solutions.
- # Collaborated with cross-functional teams to customize solutions based on client requirements, including real-time vehicle classification and LLM-based chatbot systems.
- # Leveraged state-of-the-art algorithms and techniques in computer vision, NLP, and predictive analytics for diverse application.

IIITDMJ #Jabalpur, #India
Junior Research Fellow #Jan-01 #2023 — Sep-2023

- # Conducted research on botnet detection using machine learning techniques, focusing on solutions tailored for small, resource-constrained integrated devices.
- # Designed lightweight machine learning models optimized for efficient real-time detection of malicious activities in network traffic.
- # Engineered and selected critical features from network data to enable accurate detection while minimizing computational overhead.
- # Developed and tested scalable frameworks for deployment on embedded systems, ensuring compatibility with IoT and edge devices.
- # Contributed to advancing the state of botnet detection by presenting innovative approaches at academic forums and preparing manuscripts for publication.

PROJECTS

Resume Categorization Using Machine Learning Remote, Freelance *Machine Learning Engineer* Aug 2024 — Sep 2024

- Developed a machine learning pipeline to categorize resumes into predefined job categories using TF-IDF vectorization and a Random Forest classifier.
- Processed and cleaned text data, ensuring efficient feature extraction and model performance for accurate classification.
- Automated the categorization process to organize resumes into designated folders and generate a summary CSV with filenames and predicted categories.
- Fine-tuned hyperparameters and conducted evaluation to achieve high accuracy on training, validation, and test datasets.
- Delivered a scalable and user-friendly solution for clients, enabling efficient resume organization and searchability.

PUBLICATIONS

Journal paper

An Ensemble Learning Model for Hardware Trojan Detection in Integrated Circuit Design, reviewed

- Podder, S. (2024). An Ensemble Learning Model for Hardware Trojan Detection in Integrated Circuit Design. SSRN Electronic Journal. Available at: <https://doi.org/10.2139/ssrn.4884559>

SELECTED COURSES

Master's Courses

- Deep Learning
- Data Clustering
- Research Methodology
- Hardware Security
- Embedded System
- Mobile and wireless Network

Bachelor's Courses

- Programming language 1(c)
- PROGRAMMING language 2 (c++)
- Data Structure and AlgorithmS
- Theory of Computation
- Human Computer Interaction
- Object Oriented Analysis and Design
- Computer Vision

Additional Courses

- AC,DC, Device, Dec, E-Shop
- Statistics, Math 1 to 5

AWARDS

Study in India Scholarship

#Jabalpur, #India

#Awarded the prestigious Study in India Scholarship based on outstanding academic performance, providing financial support for tuition fees and living expenses. Successfully participated in the PRAGATI exam, which evaluates skills in verbal reasoning, quantitative aptitude, and logical reasoning, as part of the scholarship selection process. Selected from a competitive pool of international students, recognized for the potential to contribute to global technological innovation. #June-07 #2021

OTHER EXPERIENCES

Teaching Assistant, Introduction to AI

#Jabalpur, #India

Teaching Assistant

#Nov-01 #2021 — #Aug-27 #2023

- Assisted with lectures and grading for an AI course, providing student guidance on core concepts.
- Organized study sessions to enhance understanding of AI algorithms and techniques.
- Designed and administered quizzes, evaluated student performance, and provided constructive feedback to support learning.

SKILLS

- **Programming:** Python, C++, SQL, TensorFlow, PyTorch, Keras, Scikit-learn
- **Software:** Jupyter Notebook, VS Code, Git, Docker, PostgreSQL, Microsoft Office Suite
- **Soft Skills:** Communication, Problem-solving, Teamwork, Critical thinking, Adaptability, Time management

REFERENCES

Prof. Aparajita Ojha

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Scholar Profiles: University of IIITDMJ - Personal Page — Google Scholar — LinkedIn

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